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(54) **THERMAL FRAME WITH INSULATING BACKING MEMBER**

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This patent is subject to a terminal disclaimer.

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(52) **U.S. Cl.**
CPC **A47F 3/0426** (2013.01)

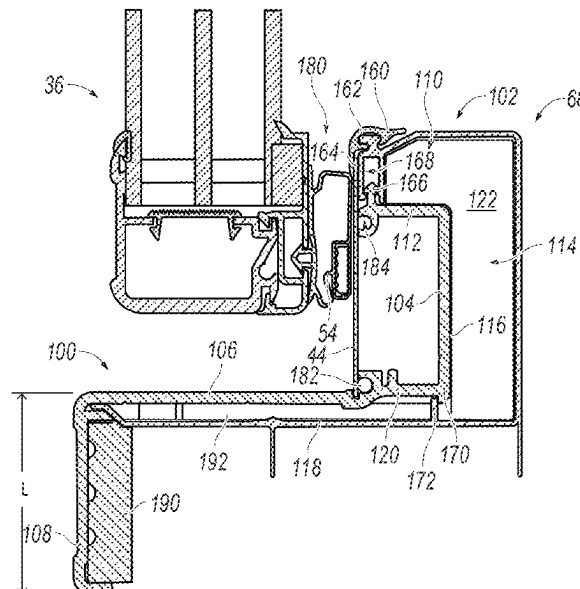
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F25D 23/082; F25D 23/085; F25D
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See application file for complete search history.

(57) **ABSTRACT**

A frame for a refrigerated enclosure including a main frame member, a backing member, and a contact plate. The main frame member includes a base, a middle wall, and a forward flange. The backing member and the contact plate are coupled to the main frame member. The backing member includes a rear leg that couples on the rear wall of the base of the main frame member and an interior leg that couples on the interior wall of the base of the main frame member. The interior leg and the rear leg each include a thermal insulating portion. The interior leg includes a front portion that couples with the contact plate. The thermally insulating portions of the interior leg and the rear leg can be integral to one another and form an L-shape.

19 Claims, 16 Drawing Sheets



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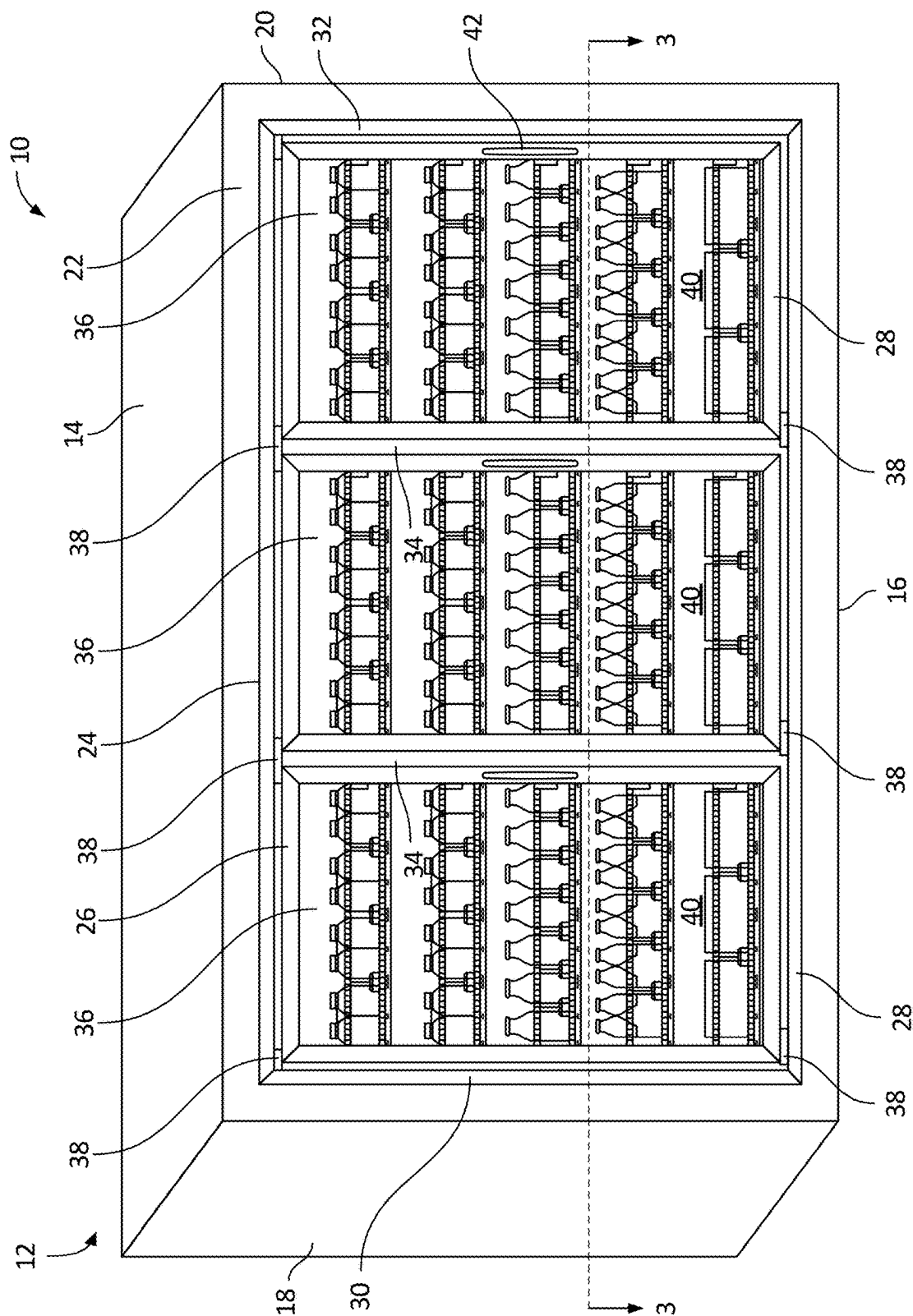


FIG. 1

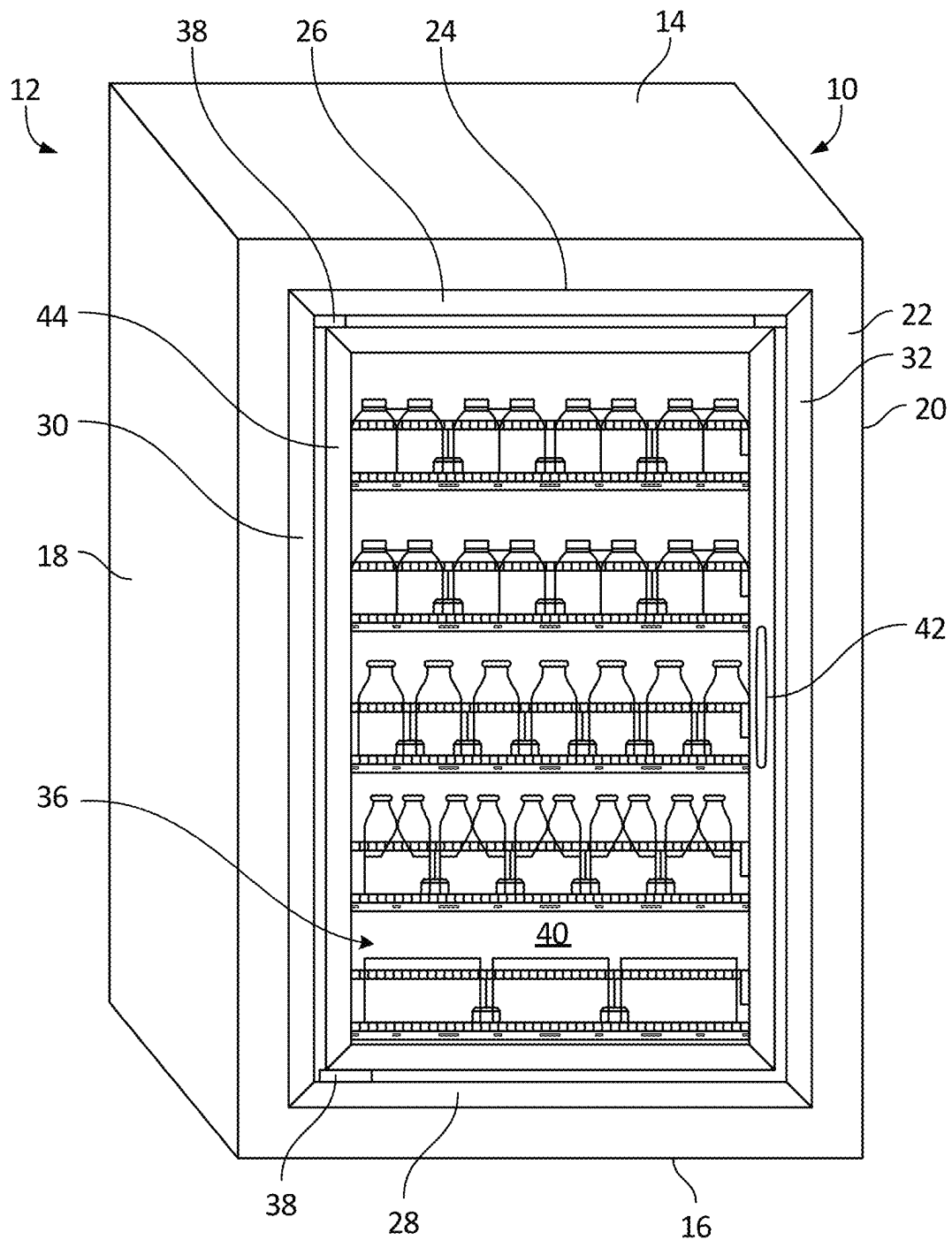


FIG. 2

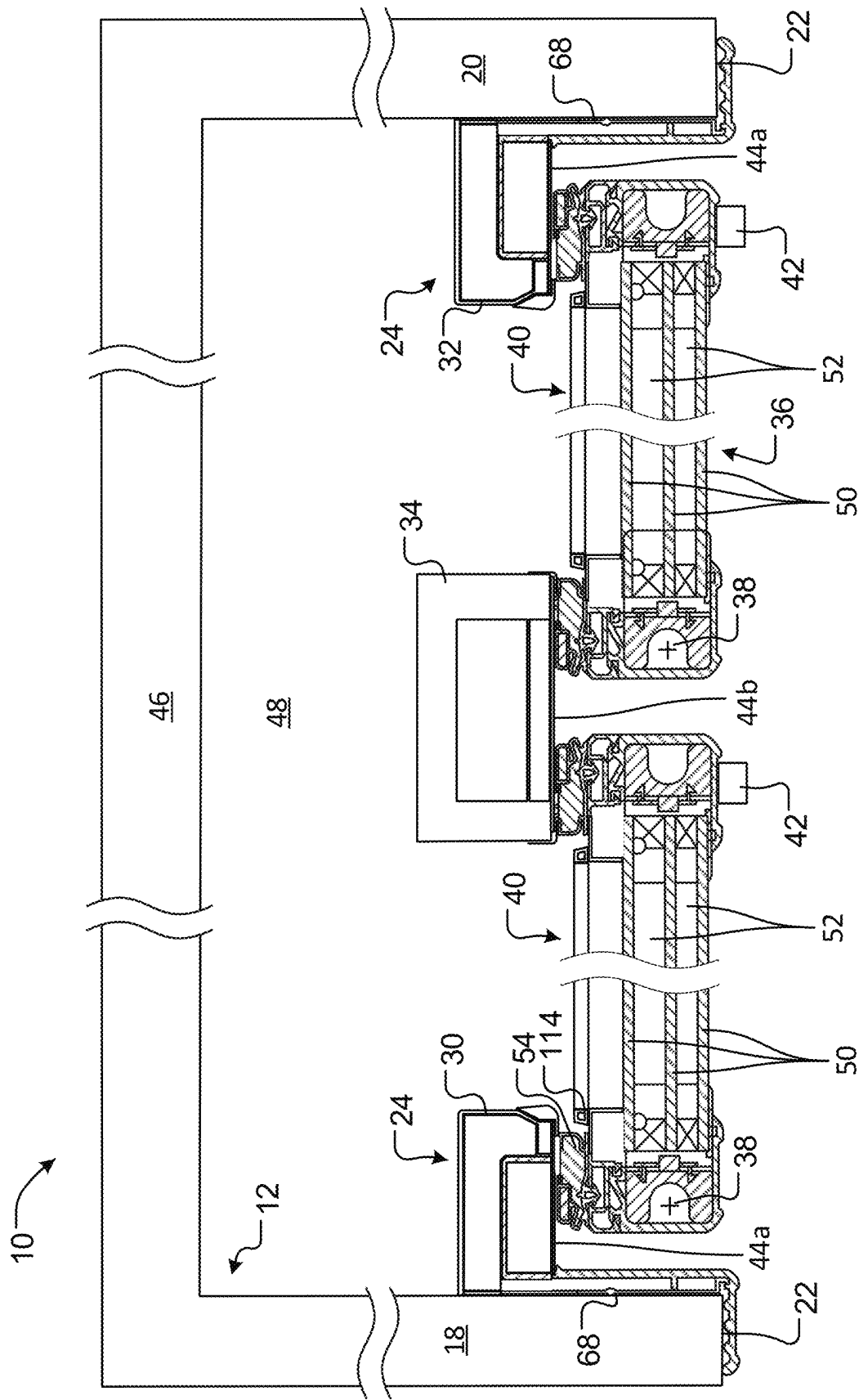
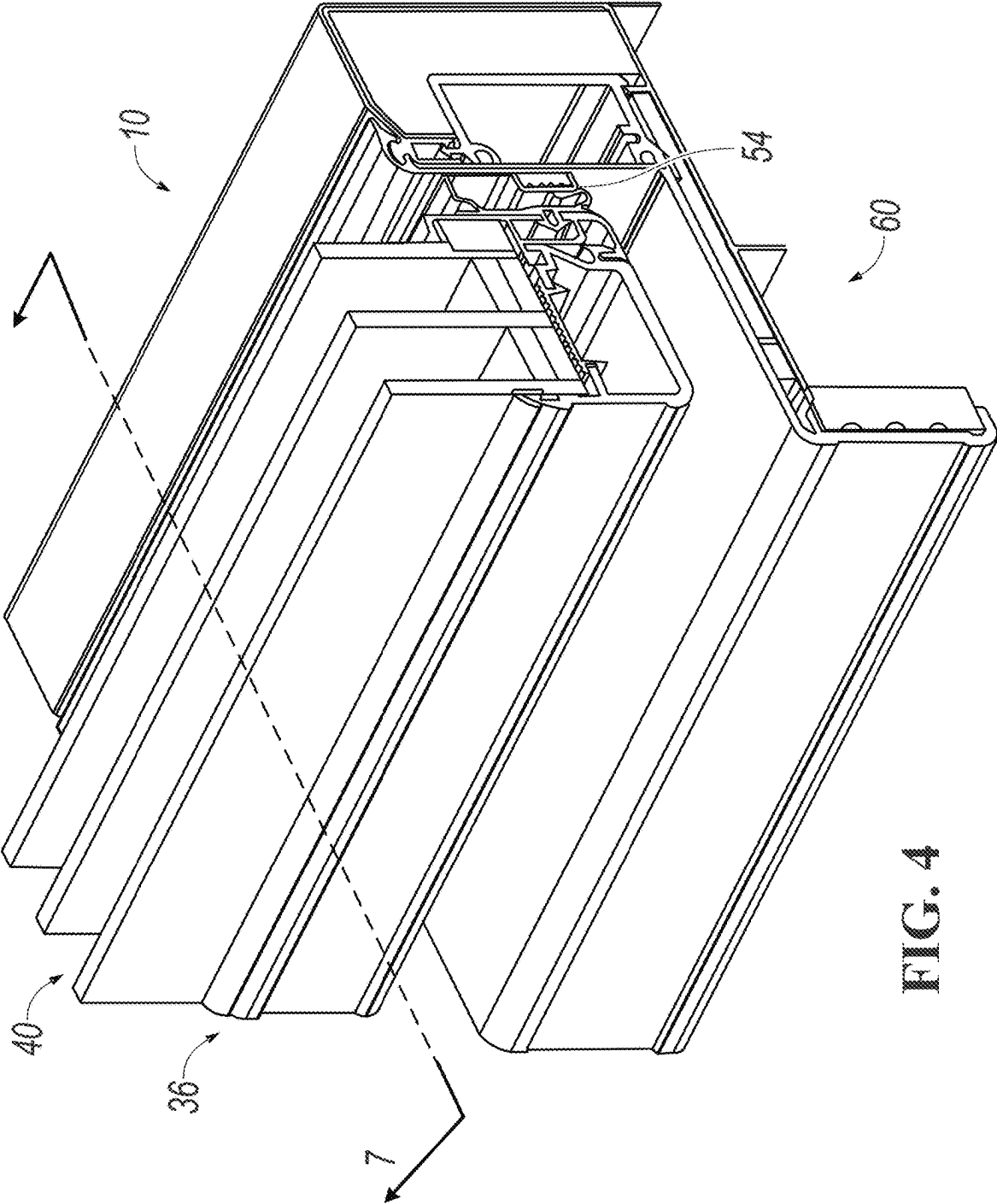
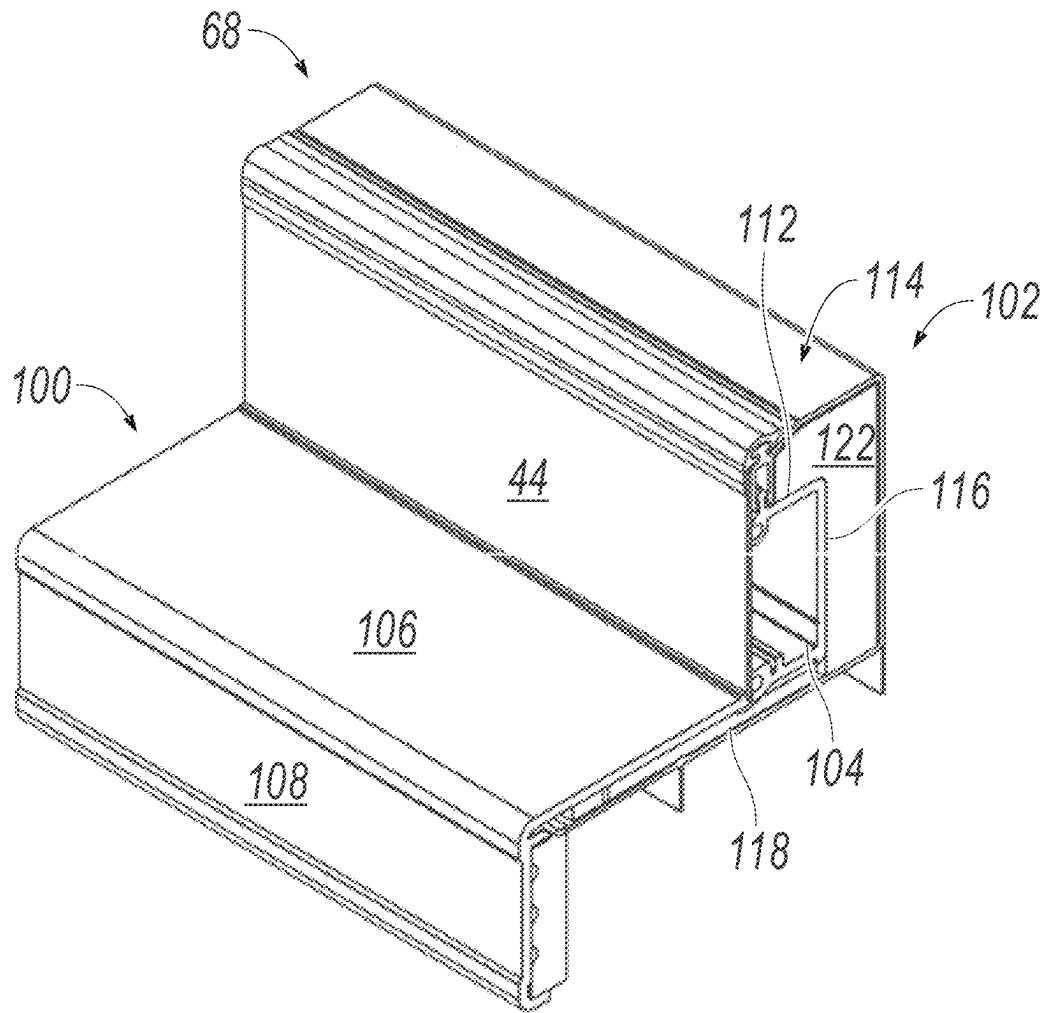
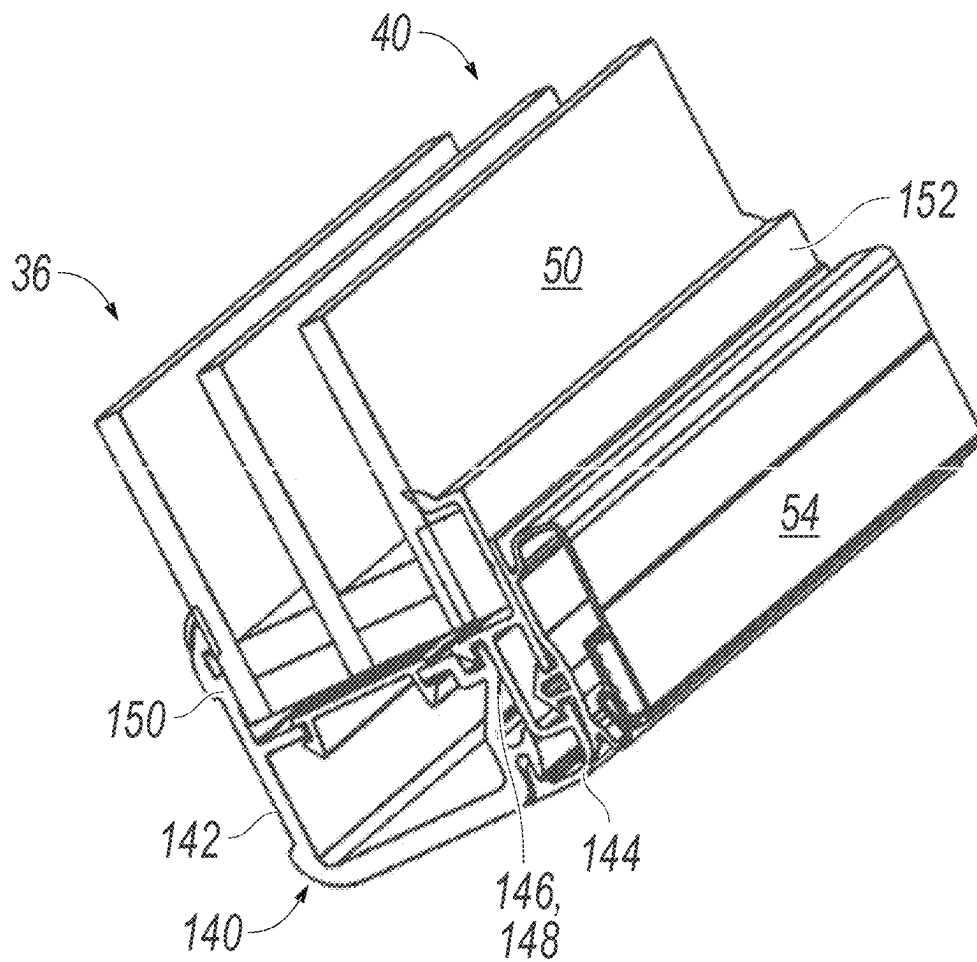


FIG. 3



**FIG. 5**

**FIG. 6**

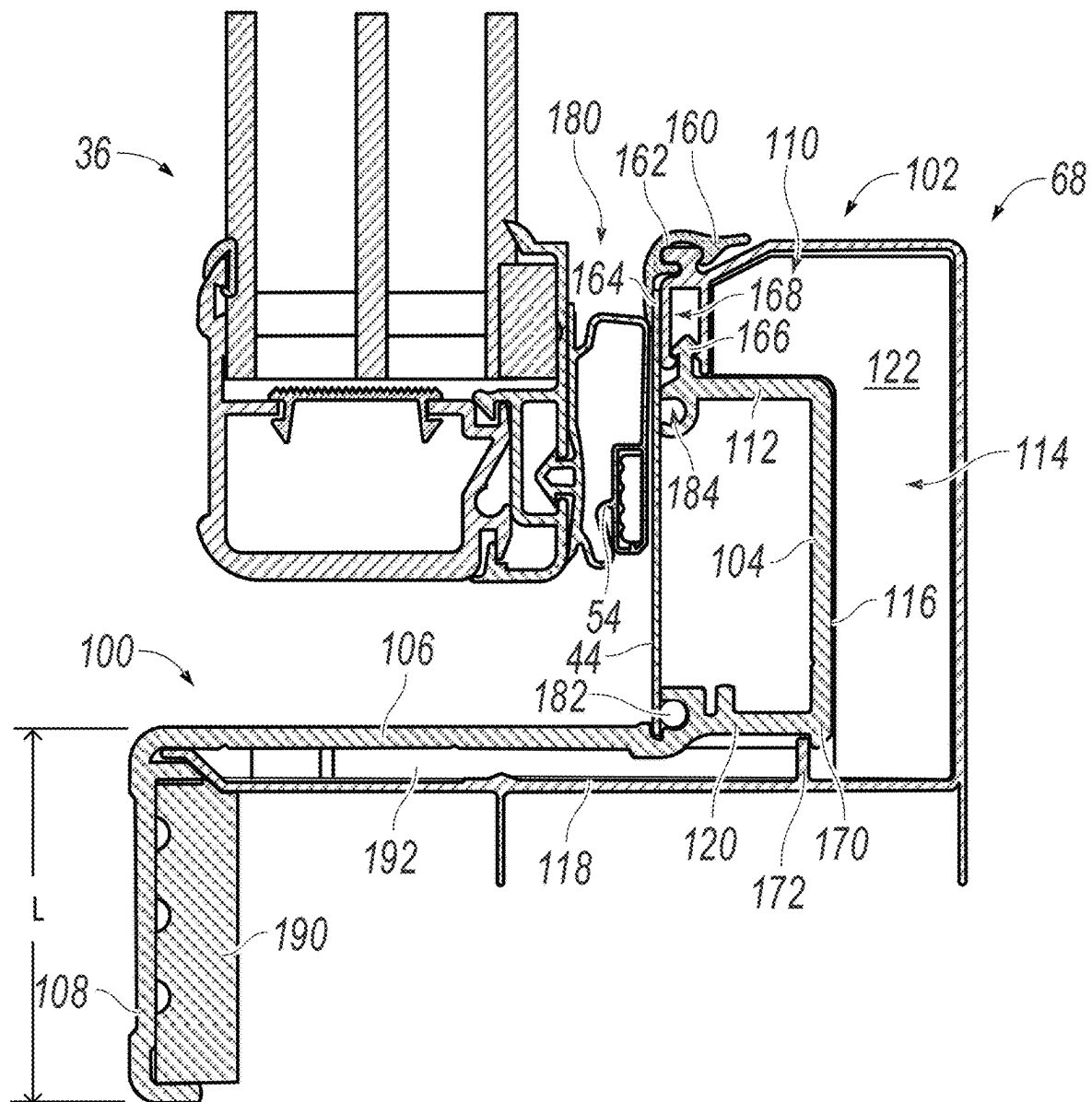


FIG. 7

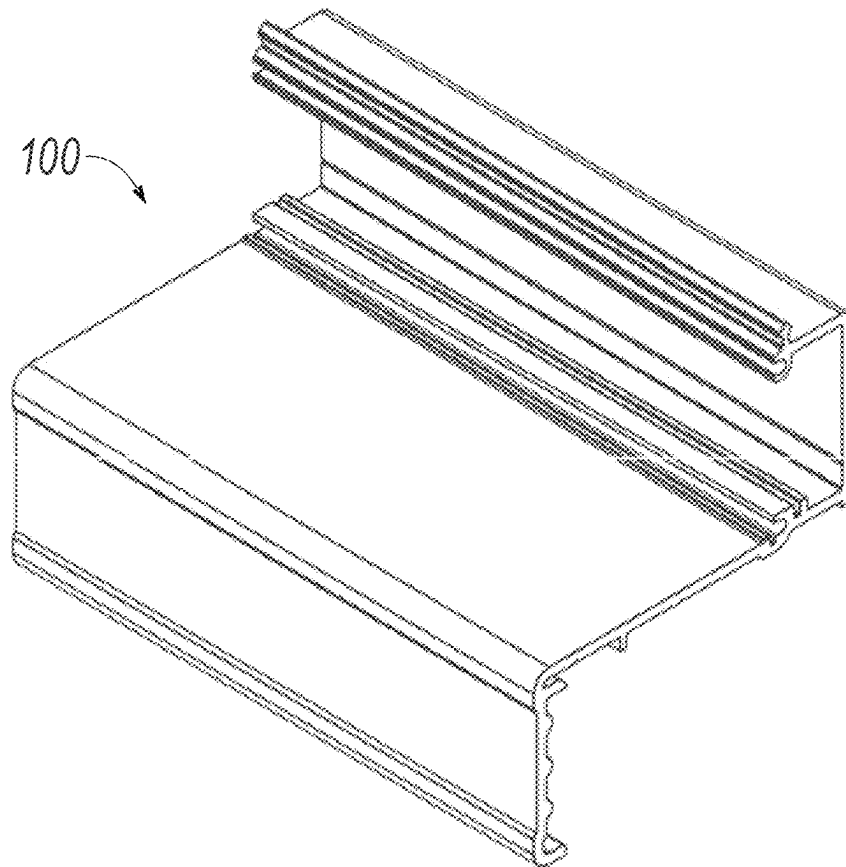
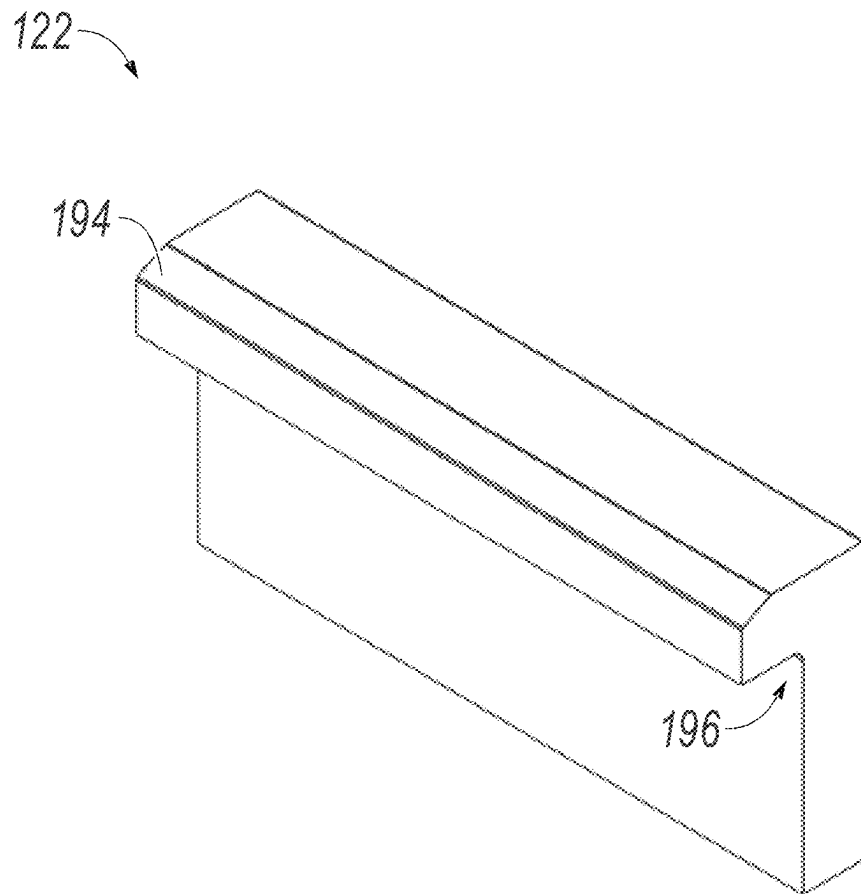


FIG. 8

**FIG. 9**

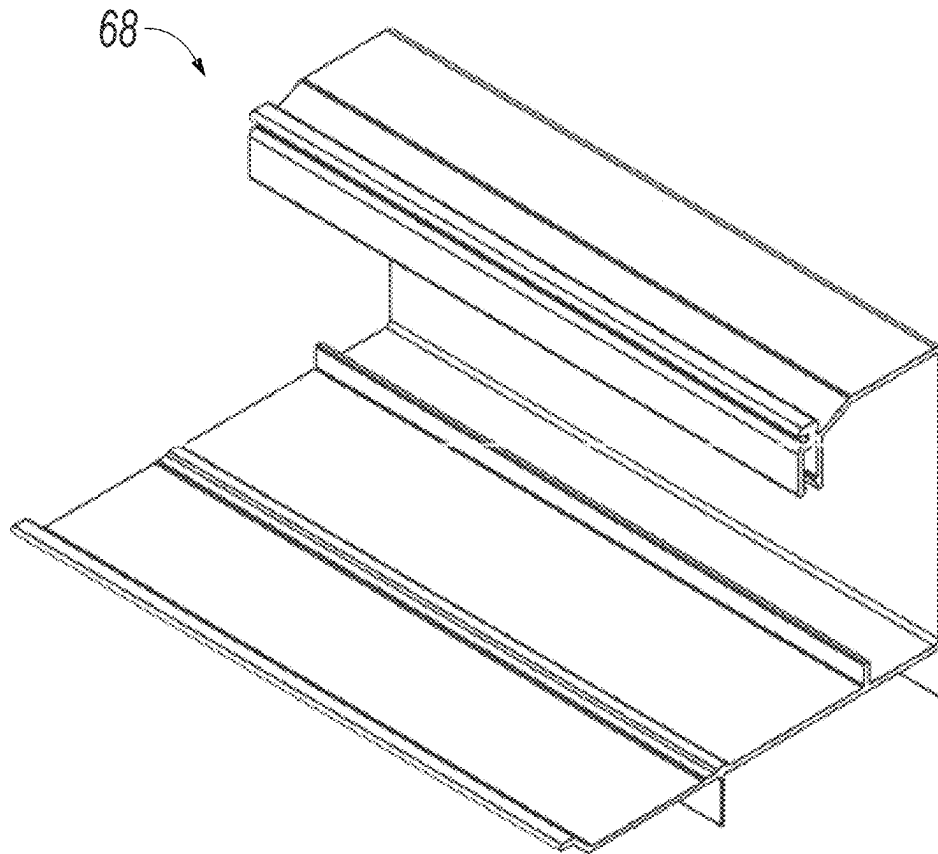


FIG. 10

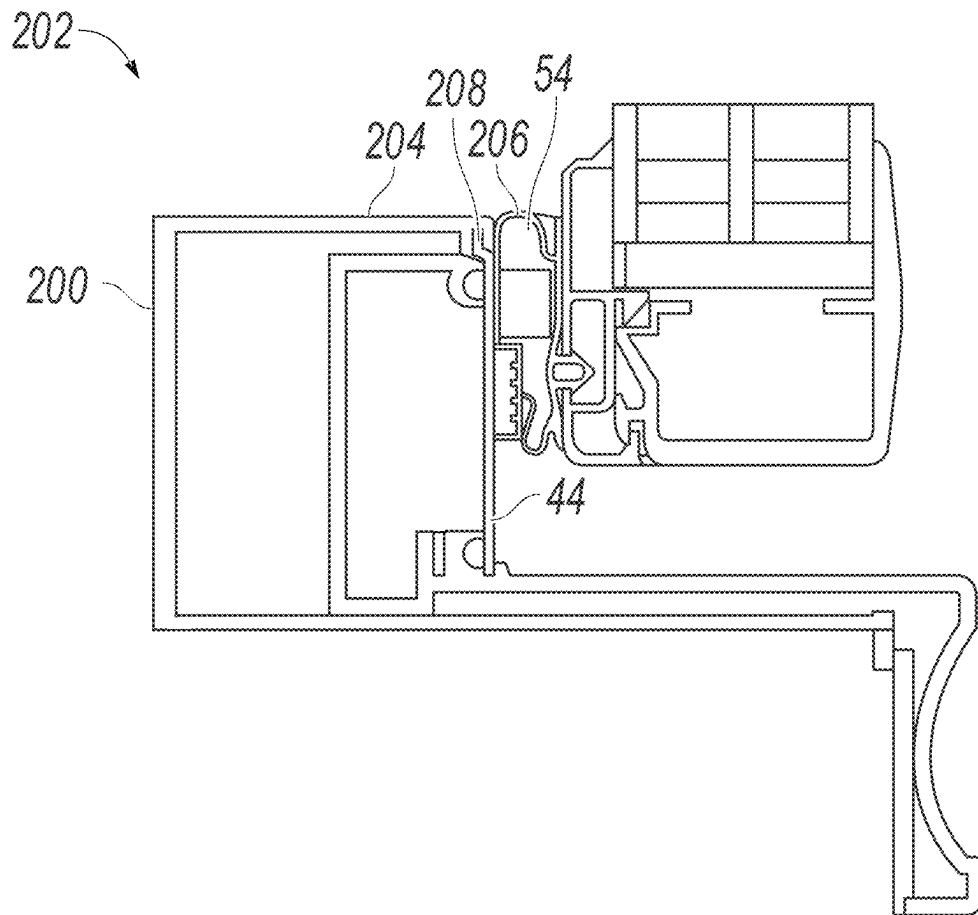


FIG. 11

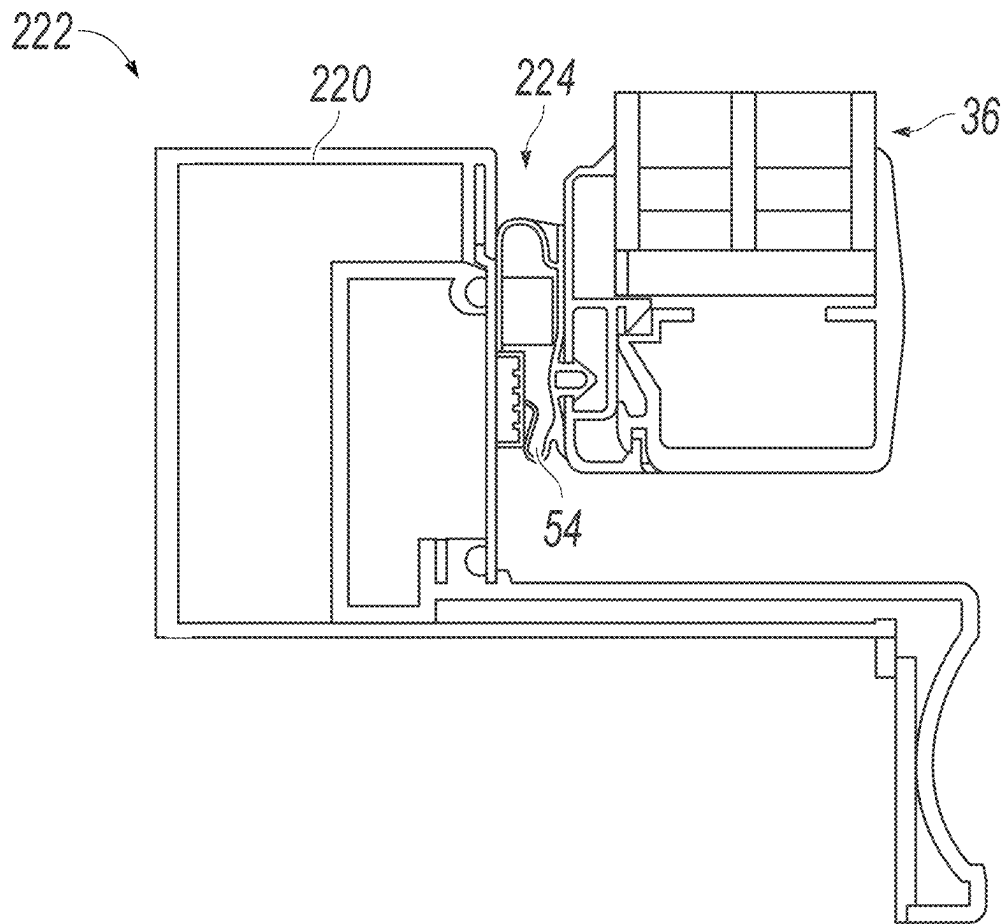


FIG. 12

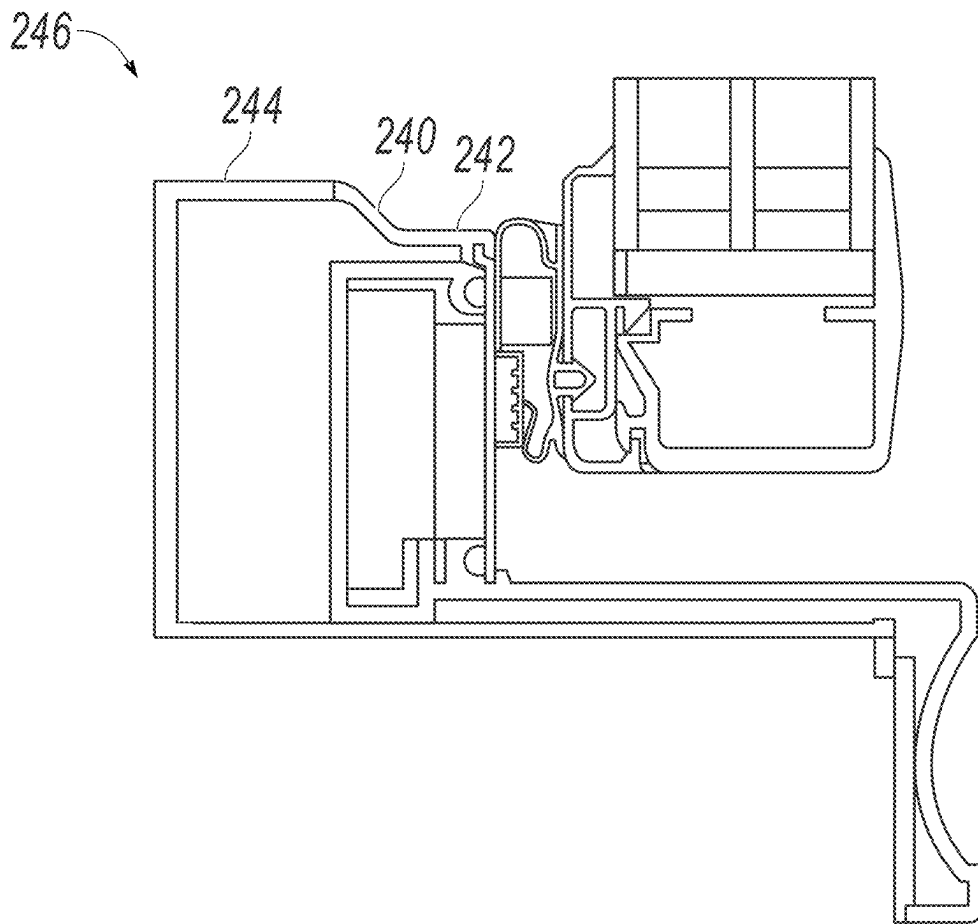


FIG. 13

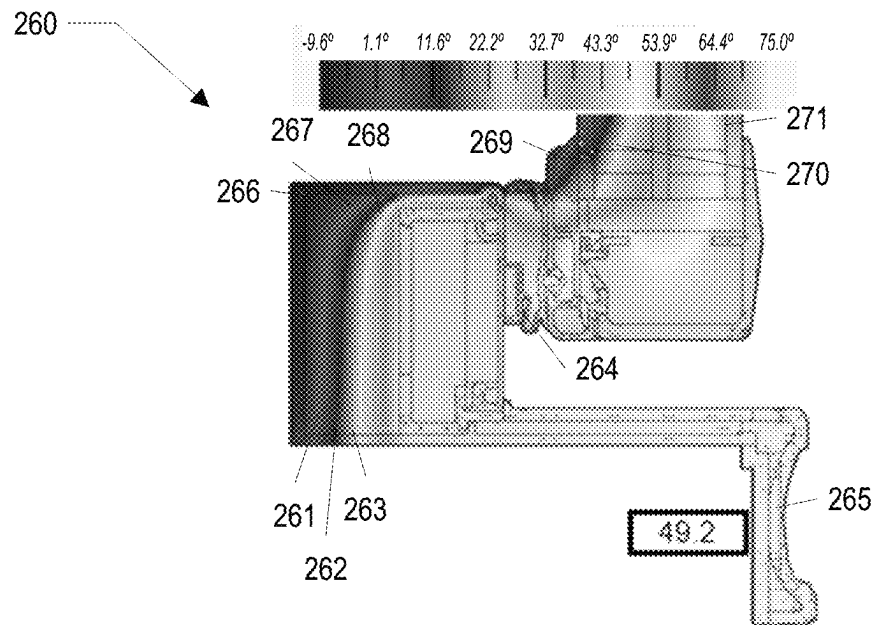


FIG. 14

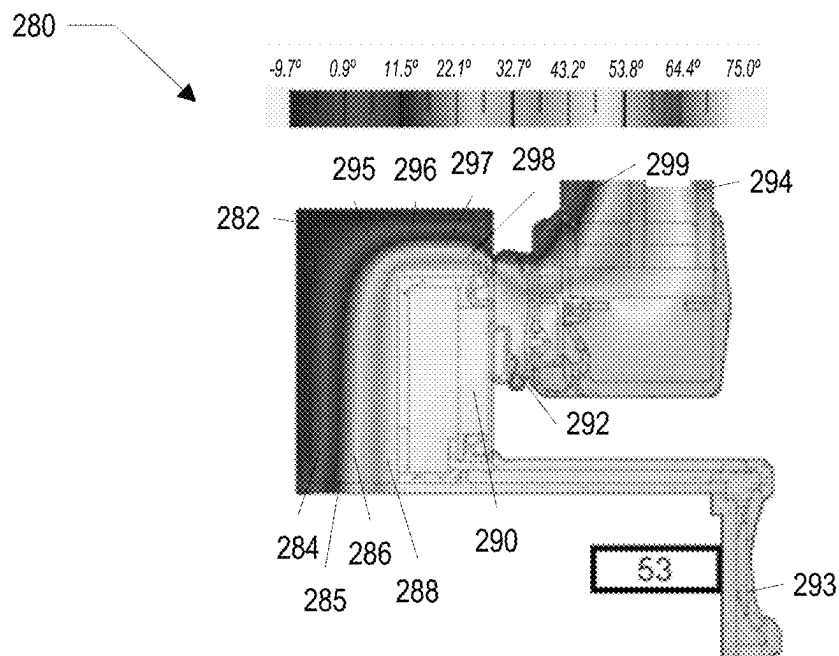


FIG. 15

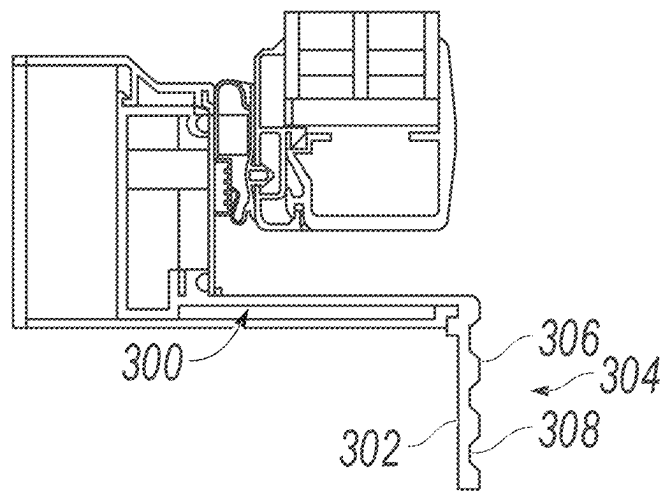


FIG. 16

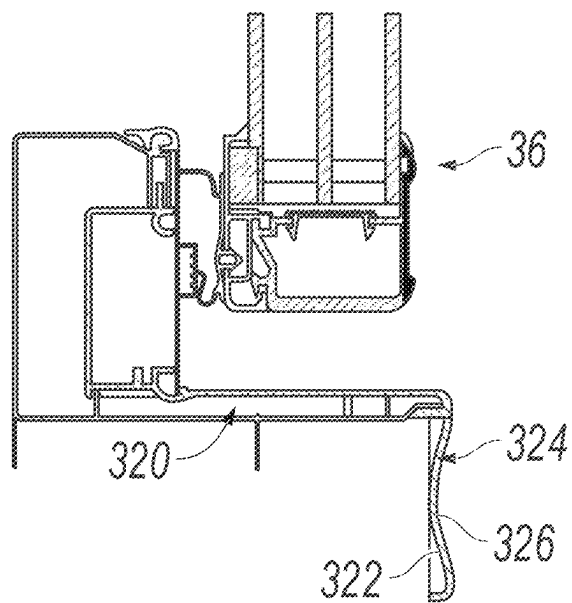
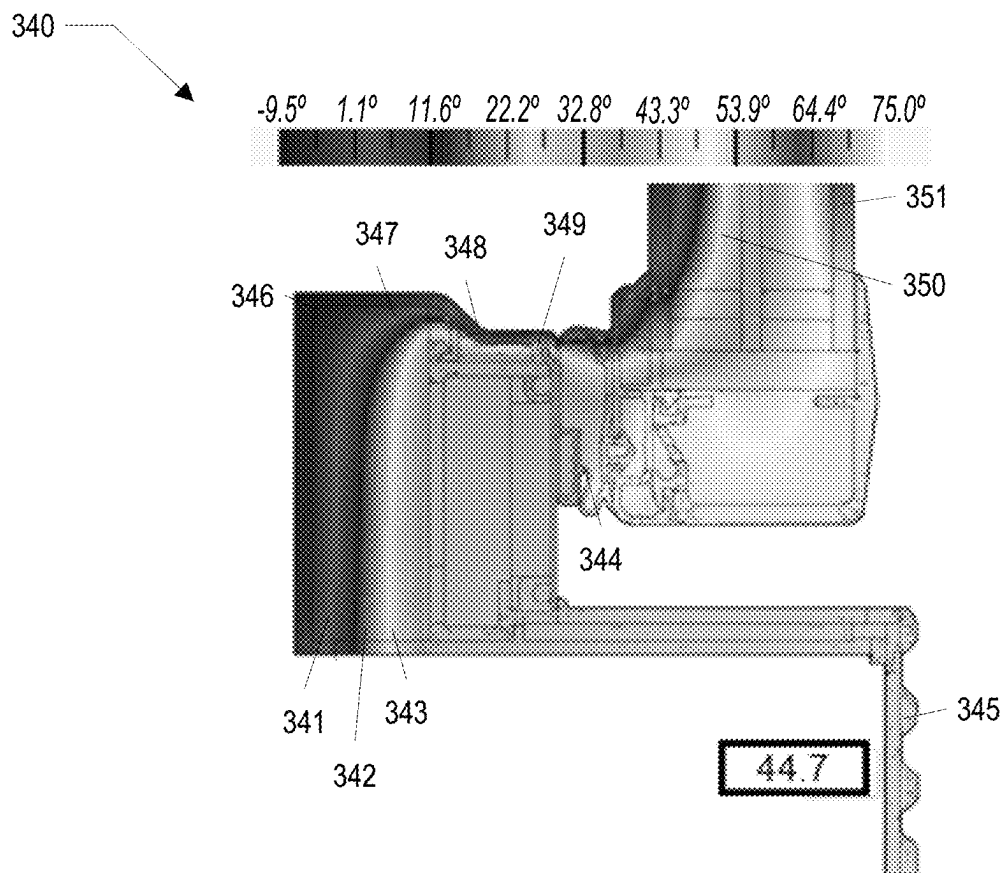


FIG. 17

**FIG. 18**

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THERMAL FRAME WITH INSULATING BACKING MEMBER

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation application of and claims priority under 35 U.S.C. § 120 to U.S. application Ser. No. 17/327,643, filed on May 21, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

This invention relates to temperature controlled storage devices, and doors and associated frames used in such devices.

BACKGROUND

Refrigerated enclosures are used in commercial, institutional, and residential applications for storing and/or displaying refrigerated or frozen objects. Refrigerated enclosures may be maintained at temperatures above freezing (e.g., a refrigerator) or at temperatures below freezing (e.g., a freezer). Refrigerated enclosures have one or more doors or windows for accessing and viewing refrigerated or frozen objects within a temperature-controlled space. Refrigerated enclosures typically include a frame that supports the doors or windows.

Condensation on sealing surfaces of doors of refrigerated enclosures and their associated frames can impair sealing and decrease energy efficiency. Formation of condensation (frost formation) on door also affects visibility to product placed inside enclosure and may cause customer dissatisfaction. Electric heater wires are sometimes employed in the thermal frames of commercial refrigerated enclosures to inhibit condensation. However, electrical heaters can use a significant amount of electrical power. Excess reliance on such heater wires may make ever more stringent government regulations on energy efficiency more difficult to meet.

SUMMARY

One aspect of the invention features a frame for a refrigerated enclosure including a main frame member, a backing member, and a contact plate. The main frame member includes a base, a middle wall, and a forward flange. The base has rear wall and an interior wall. The backing member and the contact plate are coupled to the main frame member. The backing member includes a rear leg that couples on the rear wall of the base of the main frame member and an interior leg that couples on the interior wall of the base of the main frame member. The interior leg and the rear leg each include a thermal insulating portion. The interior leg includes a front portion that couples with the contact plate.

In some implementations, the thermally insulating portions of the interior leg and the rear leg are integral to one another and form an L-shape.

In some implementations, the interior leg retains an edge of the contact plate.

In some implementations, the front portion of the interior leg extends forward to a front plane of the contact plate.

In some implementations, the thermally insulating portions of the rear leg and the interior leg include a foam material.

In some implementations, the backing member inhibits condensation on one or more surfaces of the frame.

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In some implementations, a portion of the contact plate extends in an interior direction over at least a portion of the interior leg of the backing member.

In some implementations, the frame includes a zipper coupled to the interior leg of the backing member.

In some implementations, the backing member further includes a bracket that holds the thermally insulating portions of the backing member on the base of the main frame member.

In some implementations, the bracket of the backing member and the base of the main frame member each include an engaging portion on the interior side of the base. The engaging portions engage one another to couple the backing member to the base.

In some implementations, the backing member includes an exterior leg, wherein the exterior leg is configured to extend forward from the rear leg along an exterior surface of the base of the main frame member.

In some implementations, the frame includes a thermally insulating member between the main frame member and the exterior leg of the backing member.

In some implementations, the backing member includes a bracket having an exterior portion that couples to a body of the refrigerated enclosure. The base frame member couples to the exterior portion of the bracket.

In some implementations, the base includes one or more channels that hold one or more heater wires. The channel(s) include an opening at least partially facing the contact plate.

In some implementations, the forward flange is configured to absorb heat from ambient air to inhibit condensation on the frame.

Another aspect of the invention features a temperature-controlled enclosure for displaying cold items. The temperature-controlled enclosure includes a body, a frame assembly, and one or more doors. The body includes a front opening and defines an interior space of the enclosure. The frame assembly is coupled in the front opening of the body. The frame assembly includes a frame segment having, in cross-section, a main frame member, a backing member, and a contact plate. The rear leg couples on the rear wall of the base of the main frame member, and the interior leg couples on the interior wall of the base of the main frame member. The interior leg and the rear leg each comprise a thermally insulating portion. A front portion of interior leg of the backing member extends forward to a front plane of the contact plate. The gasket forms a seal between the frame assembly and the door when the door is closed. A portion of the backing member of the frame segment contacts at least a portion of a rear surface of the gasket or a rear surface of the contact plate when the door is closed.

In some implementations, an inner surface of the interior leg of the backing member is interior to an inner surface of the gasket when the door is closed.

In some implementations, an inner surface of the backing member is substantially aligned with an inner surface of the gasket when the door is closed.

In some implementations, a front surface of the backing member and a rear surface of the door define a channel interior to an inner surface of the gasket.

In some implementations, the forward flange is configured to absorb heat from the ambient air to inhibit condensation on the frame.

In some implementations, the temperature-controlled enclosure includes an insulating member between the forward flange and a front surface of the body of the enclosure.

Another aspect of the invention features a frame for a refrigerated enclosure. The frame includes a frame segment

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having, in cross-section, a main frame member, a backing member, and a contact plate. The main frame member includes a base having a rear wall and an interior wall.

The backing member includes a rear leg that couples on the rear wall of the base of the main frame member, and an interior leg that couples on the interior wall of the base of the main frame member. The interior leg and the rear leg each include a thermal insulating portion. The thermally insulating portions of the interior leg and the rear leg form an L-shape. The contact plate extends inwardly over at least a portion of the front portion of the backing member.

The concepts described herein may provide several advantages. For example, implementations of the invention may provide a frame with improved thermal efficiency. Implementations may prevent or minimize condensation build up on door sealing surfaces. Implementations may provide for a more positive thermal seal between a thermal frame and a door.

The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view of a refrigerated enclosure having multiple doors supported by a frame.

FIG. 2 is a perspective view of a refrigerated enclosure having a single door supported by a frame.

FIG. 3 is a cross-sectional view showing an example refrigerated enclosure with two doors and a mullion according to implementations of the present disclosure.

FIG. 4 illustrates refrigerated enclosure with a door in a closed position on a frame assembly according to implementations of the present disclosure.

FIG. 5 is a perspective view of a frame segment assembly according to an illustrative implementation.

FIG. 6 is a perspective view of a door according to an illustrative implementation.

FIG. 7 is a cross-sectional view of a door in a closed position on a frame assembly according to an illustrative implementation.

FIG. 8 depicts an example of a main frame member according to an illustrative implementation.

FIG. 9 depicts an example of an insulating member according to an illustrative implementation.

FIG. 10 depicts an example of a bracket of a backing member according to an illustrative implementation.

FIG. 11 depicts an example of a frame segment including a backing member that is in direct contact with a door gasket.

FIG. 12 depicts another example of a frame segment assembly including a backing member.

FIG. 13 depicts an example of a backing member including a step.

FIG. 14 shows a thermal map of results from thermal modeling performed on a door and frame segment assembly shown in FIG. 11.

FIG. 15 shows a thermal map of results from thermal modeling performed on a door and frame segment assembly shown in FIG. 12.

FIG. 16 illustrates a thermal frame having a forward ball flange according to implementations of the present disclosure.

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FIG. 17 illustrates a thermal frame having a forward flange with a curved surface according to implementations of the present disclosure.

FIG. 18 shows a thermal map of results from thermal modeling performed on the door and frame segment assembly of FIG. 16.

Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

In some implementations, a frame segment assembly includes an L-shaped backing member that fits on the back and interior faces of an inner member of the frame. The backing member includes a thermally insulating member for reducing thermal transference between the frame and the interior space of the enclosure. The insulating member can be an L-shape. In some implementations, the interior leg of the L-shaped backing member runs from the back of the frame to a rear surface of the door gasket. The contact plate of the frame can extend over the interior leg of the backing member.

FIGS. 1-2 show an exemplary refrigerated enclosure 10. Refrigerated enclosure 10 may be a refrigerator, freezer, or other enclosure defining a temperature-controlled space. In some implementations, refrigerated enclosure 10 is a refrigerated display case. For example, refrigerated enclosure 10 may be a refrigerated display case or refrigerated merchandiser in grocery stores, supermarkets, convenience stores, florist shops, and/or other commercial settings to store and display temperature-sensitive consumer goods (e.g., food products and the like). Refrigerated enclosure 10 can be used to display products that must be stored at relatively low temperatures and can include shelves, glass doors, and/or glass walls to permit viewing of the products supported by the shelves. In some implementations, refrigerated enclosure 10 is a refrigerated storage unit used, for example, in warehouses, restaurants, and lounges. Refrigerated enclosure 10 can be a free standing unit or “built in” unit that forms a part of the building in which refrigerated enclosure 10 is located.

Refrigerated enclosure 10 includes a body 12. Body 12 includes a top wall 14, a bottom wall 16, a left side wall 18, a right side wall 20, a rear wall (not shown), and a front portion 22 defining a temperature-controlled space. Front portion 22 includes an opening into the temperature-controlled space. Thermal frame 24 is can be mounted at least partially within the opening. Thermal frame 24 includes a plurality of perimeter frame segments (i.e., a header or top frame segment 26, a sill or bottom frame segment 28, a left side frame segment 30, and a right side frame segment 32) forming a closed shape along a perimeter of the opening. In some implementations, thermal frame 24 includes one or more mullion frame segments 34 dividing the opening into multiple smaller openings. For example, FIG. 1 illustrates a three-door assembly with a pair of mullion frame segments 34 extending between top frame segment 26 and bottom frame segment 28 to divide the opening into three smaller openings. Each of the smaller openings may correspond to a separate door 36 of the three-door assembly. In other implementations, mullion frame segments 34 may be omitted. For example, FIG. 2 illustrates a one-door assembly wherein thermal frame 24 includes perimeter frame segments 26-32 but not mullion frame segments 34. In some implementations, thermal frame 24 includes include top frame segment 26 and bottom frame segment 28 with no side frame segments 30 or 32. In such implementation, thermal

frame 24 may include one or more mullion frame segments 34 depending, for example, on the size of the refrigerated enclosure in which thermal frame 24 is to be installed and the number of doors.

Refrigerated enclosure 10 includes one or more doors 36 pivotally mounted on the thermal frame 24 by hinges 38. In some implementations, the doors 36 are sliding doors configured to open and close by sliding relative to the thermal frame 24. The example doors 36 illustrated in FIGS. 1 and 2 include panel assemblies 40 and handles 42. Referring to FIG. 2, thermal frame 24 includes a series of contact plates 44. Contact plates 44 are attached to a front surface of thermal frame 24 and provide a sealing surface against which doors 36 rest in the closed position. For example, doors 36 may include a gasket or other sealing feature around a perimeter of each door 36. The gaskets may employ a flexible bellows and magnet arrangement, which, when the doors 36 are closed, engage contact plates 44 to provide a seal between doors 36 and thermal frame 24. The thermal frames provide a thermally conductive path from the frame segments 26-32, for maintaining maintains the temperature of the contact plates 44 at or close to the temperature of the external environment (the environment outside of the refrigerated enclosure 10) and to aid in preventing condensation from forming on the contact plates 44. Preventing condensation on the contact plates may provide for a more positive seal between the contact plates 44 and a magnetic gasket on the door, thereby improving the thermal properties of the refrigerated enclosure 10.

FIG. 3 illustrates a cross-sectional view of the refrigerated enclosure 10 taken along the line 3-3 in FIG. 1. FIG. 3 illustrates the pair of side walls 18 and 20 of the refrigerated enclosure 10 extending rearward from front portion 22, and a rear wall 46 extending between side walls 18 and 20 to define a temperature-controlled space 48 within the body 12.

In FIG. 3, refrigerated enclosure 10 is shown as a two-door assembly with a pair of doors 36 positioned in an opening in front portion 22. Refrigerated enclosure 10 may have two doors 36 (as shown in FIG. 3), a lesser number of doors 36 (e.g., a single door as shown in FIG. 2), or a greater number of doors 36 (e.g., three or more doors as shown in FIG. 1). Each door 36 includes a panel assembly 40 and a handle 42. Applying a force to handle 42 causes the corresponding door 36 to rotate about hinges 38 between an open position and a closed position. In some implementations, panel assembly 40 is a transparent or translucent panel assembly through which items within temperature-controlled space 48 can be viewed when doors 36 are in the closed position. For example, panel assembly 40 is shown to include a plurality of transparent or translucent panels 50 with spaces 52 therebetween. The spaces 52 can be sealed and filled with an insulating gas (e.g., argon) or evacuated to produce a vacuum between panels 50. In some embodiments, panel assembly 40 includes opaque panels with an insulating foam or other insulator therebetween. Doors 36 include gaskets 54 attached to a rear surface of doors 36 along an outer perimeter of each door. Gaskets 54 are configured to engage a sealing surface of the contact plates 44a and 44b (referred to collectively as contact plates 44) when the doors 36 are in the closed position, and to thereby provide a seal between doors 36 and contact plates 44.

The perimeter frame segments 30-32 of the thermal frame 24 are coupled to the body 12 of the refrigerated enclosure 10 by mounting brackets 68. Mounting brackets 68 can be secured to perimeter frame segments 30-32 using one or more connection features (e.g., flanges, notches, grooves, collars, lips, etc.) or fasteners (e.g., bolts, screws, clips, etc.)

and may hold perimeter frame segments 30-32 in a fixed position relative to the body 12 of the refrigerated enclosure 10.

Although only two perimeter frame segments 30-32 are shown in FIG. 3, other perimeter frame segments (e.g., header/top frame segment 26 and sill/bottom frame segment 28) may be configured in a similar manner. For example, top frame segment 26 and bottom frame segment 28 may be coupled to the body 12 of the refrigerated enclosure 10 by mounting brackets 68.

The perimeter frame segment assembly includes a perimeter frame segment (i.e., one of frame segments 26-32), a mounting bracket 68, and a contact plate 44.

One or more mullion frame segments 34 extend vertically between top frame segment 26 and bottom frame segment 28. A top portion of mullion frame segment 34 is fastened to a top frame segment 26 and a bottom portion of mullion frame segment 34 is fastened to a bottom frame segment 28.

In some implementations, the frame assembly includes an L-shaped thermally insulating backing member that fits on the back and interior faces of the main frame member of a mounting frame for the door of a commercial refrigerated enclosure. The backing member includes insulation for reducing thermal transference between the frame and the interior space of the enclosure. The interior leg of the L-shaped backing member may run from the back of the frame to the trailing edge of the door gasket. The contact plate of the frame can extend an interior direction over the interior leg of the backing member.

FIG. 4 illustrates refrigerated enclosure with a door in a closed position on a frame assembly according to implementations of the present disclosure. Refrigerated enclosure 10 includes door 36 and frame segment assembly 60. Door 36 is closed on frame segment assembly 60. For illustrative purposes, only a short segment of the door and a corresponding segment of frame segment assembly shown. Nevertheless, the cross section shown in FIG. 4 can continue around the entire perimeter of window panel assemblies 40 and the corresponding perimeter of the frame assembly.

Door 36 includes window panel assembly 40 and gasket 54. Gasket 54 may run continuously around the perimeter of door 36. In various implementations, gasket 54 can be a single continuous piece, or can include a set of gasket components, with one gasket component on each of the edges of the perimeter. Gasket 54 can be made of a resilient material, such as synthetic rubber.

FIG. 5 is a perspective view of a frame segment assembly according to an illustrative implementation. Frame segment assembly 60 includes main frame member 100, backing member 102, and contact plate 44. Main frame member 100 includes base 104, middle wall 106, and forward flange 108.

Backing member 102 is coupled to main frame member 100. Interior leg 110 of backing member 102 is against interior wall 112 of main frame member 100. Rear leg 114 of backing member 102 is against rear wall 116 of main frame member 100. Exterior wall 118 of bracket 68 may run along an outer side wall 120 of base 104 and middle wall 106.

Backing member 102 includes bracket 68 and insulating member 122. In the implementation shown in FIG. 5, insulating member 122 has an L-shape with one leg of the insulating member forming part of rear leg 114 of backing member 102, and another leg of the insulating member forming a part of interior leg 110. Over a portion of exterior wall 118, insulating member 122 is disposed between exterior wall 118 of bracket 68 and the middle wall 106 of main frame member 100.

In some implementations, the interior leg of an insulating member extends all the way to a contact plate or gasket. In some implementations, the interior leg of insulating member extends more than half way from the rear edge of base **104** to the contact plate or more than half way from the rear edge of base **104** to rear surface of the gasket. In one implementation, a rear leg extends all the way across the rear surface of base **104**. In one implementation, a rear leg extends more than half way across the rear surface of base **104**. In some implementations, a rear leg extends all the way to a bracket in contact with the body of a refrigerated enclosure.

FIG. 6 is a perspective view of a door according to an illustrative implementation. Window panel assembly **40** includes panels **50** and housing assembly **140**. Housing assembly **100** surrounds and supports the edges of window panels **50**. Housing assembly **140** includes outer housing member **142** and rear retaining member **144**. Rear retaining member **144** can snap together with outer housing member **142** by way of complementary engaging portions **146**, **148**. Window panels **50** are stacked between front retaining rim **150** of outer housing member **142** and rear retaining rim **152** of rear retaining member **144**.

FIG. 7 is a cross-sectional view of a door in a closed position on a frame assembly according to an illustrative implementation. The outer end of contact plate **44** is supported by main frame member **100**. Contact plate **44** may be secured in place with a retaining clip **160** (e.g., a zipper strip or other suitable fastening device). Retaining clip **160** may be coupled to bracket **68** by an engagement feature **162** (e.g., a flange, a notch, a lip, a collar, a groove, etc.) of bracket **68** of backing member **102**.

In the example shown in FIG. 7, bracket **68** extends to the plane of the front of base **104** of main frame member **100**. Front surface **164** of bracket **68** contacts the rear surface of contact plate **44**. Bracket **68** is secured to the interior side of base **100** of main frame member **100** by way of engagement of projection **166** of base **104** in retaining groove **168** of bracket **68**.

In this implementation, insulating member **122**, the front portion of bracket **68**, and contact plate **44** form a continuous thermal barrier between temperature controlled space **48** of the refrigerated enclosure and main frame member **100**. Retaining groove **168** can be deeper than the length of projection **166** such that an air pocket is defined inside the groove when the backing member **102** is coupled with main frame member **100**.

On the exterior side of main frame member **100**, projecting rim **170** of base **102** can engage with rib **172** on exterior wall **118** of bracket **68**. In some cases, main frame member **100** and backing member **102** can be snapped into engagement with one another.

As assembled, contact plate **44** extends in an interior direction over a portion of backing member **102**.

With door **36** closed on frame segment **60**, a channel **180** is defined by the rear surface of door **36**, a front surface of the frame segment **60**, and an interior surface of gasket **54**.

In some implementations, a main frame member includes heater wire channels that position the heater wire in direct contact with the contact plate of the frame.

In the example shown in FIG. 7, heater wires **182** are received in channels **184** formed in base **104** of main frame member **100**. Channels **184** are open to the front of base **100** such that heater wires **182** can be in direct contact with contact plate **44**.

In the implementation shown in FIG. 7 contact plate **44** is sandwiched between front surface **164** of backing member **102** and a rear surface of gasket **54**. In other implementa-

tions, a backing member can be in direct contact with a door gasket when the door is closed.

In certain implementations, a portion of a backing member can project forward such that backing member overlaps the rear edge of the gasket. For example, in an alternate implementation, a portion of backing member **102** can extend into the region of channel **180**.

Insulating member **190** is interposed between forward flange **108** of main frame member **100** and body **12** of the refrigerated enclosure **10**. Insulating member **190** can be, in one example, seal tape. Insulating member **192** is interposed between exterior wall **118** of bracket **68** and middle wall **106** of main frame member **100**. Insulating member **192** can be, in one example, a foam gasket. Insulating members **190** and **192** may thermally insulate main frame member **100** from body **12** and help maintain contact plate **44** and door **36** at temperatures that inhibit condensation on contact plate **44** and surfaces of door **36**.

FIG. 8 depicts an example of a main frame member according to an illustrative implementation. Main frame member **100** can be, in some implementations, made of aluminum.

FIG. 9 depicts an example of an insulating member according to an illustrative implementation. In this example, insulating member **122** generally has an L-shape. Some surfaces of insulating member **122** can be sloped or contoured. For example, in the implementation shown in FIG. 9, insulating member **122** includes chamfer **194** and fillet **196**. Insulating member **122** can be, in some implementations, made of an extruded polystyrene foam material such as Blue Board produced by Dow Chemical Company. In certain implementations, insulating member **122** can be made of a cellular PVC foam material. In some implementations, insulating material can be a Celuka material. In some implementations, insulating member **122** can be made of polystyrene. In certain implementations, an insulating member can include a vacuum insulated panel. Other thermally insulating materials can be used in various implementations.

FIG. 10 depicts an example of a bracket of a backing member according to an illustrative implementation. Bracket **68** can be, in some implementations, made of a polystyrene.

In some implementations, a backing member for a main frame segment wraps around the main frame member such as to be in direct contact with a door gasket. FIG. 11 depicts an example of a frame segment including a backing member that is in direct contact with a door gasket. In this example, the front portion of bracket **200** of backing member **202** captures and retains the interior edge of contact plate **44**. The front edge of bracket **200** is coplanar with the front surface of contact plate **44** and contacts the rear surface of gasket **54**. In the example shown in FIG. 11, the interior face **204** of backing member **202** is generally aligned with an interior surface **206** of gasket **54**. The interior edge of contact plate **44** includes tab **208**. Tab **208** is offset to the rear of the front surface of contact plate **44**.

In some implementations, the interior edge of a backing member for a frame segment projects inwardly beyond the interior edge of a door gasket. FIG. 12 depicts another example of a frame segment assembly including a backing member. Inner surface **220** of backing member **222** is interior to the inner surface of gasket **54**. A portion of the front edge of backing member **222** and a portion of rear surface of door **36** define a channel **224** between the interior surface of gasket **54** and the temperature controlled space. In

one implementation, backing member **222** projects at least about 10 millimeters from the interior surface of gasket **54**.

In some implementations, an interior surface of a backing member varies in thickness over the length of the interior leg. FIG. **13** depicts an example of a backing member including a step. Step **240** is included between interior surface **242** and interior surface **244** of backing member **246**.

Inhibiting transfer of heat to the cold interior space at the interface of a door and frame may help maintain the temperature of the sealing surface of a contact plate above the dew point of the external environment. This inhibits condensation from forming on the sealing surface of the contact plate. Prevention of condensation on the sealing surface may promote positive engagement and improved thermal seals between contact plates and door gaskets.

FIG. **14** shows a thermal map **260** of results from thermal modeling performed on the door and frame segment assembly of FIG. **11**. As illustrated by the temperature regions extending along the outer member and to the contact plate, the thermally conductive outer member of the frame assembly readily conducts heat from the external environment to the thermal plate. Thus, main frame member **202** and contact plate **44** may be maintained at a relatively uniform temperature with the external environment. On the interior side of the frame segment, there is a relatively steep temperature gradient, as indicated by the rapid transition of temperature regions in a short distance from the interior surface of the backing member. This steep temperature gradient indicates that the backing member is preventing heat from the external environment from entering into the inside of the refrigerated enclosure. For illustrative purposes, the following are approximate temperatures at locations **261** through **271** shown in FIG. **14**: **261**: 1.0° F., **262**: 11.6° F., **263**: 22.2° F., **264**: 48.6° F., **265**: 49.2° F., **266**: -9.7° F., **267**: 1.0° F., **268**: 22.2° F., **269**: 1.0° F., **270**: 22.2° F., **271**: 53.9° F.,

FIG. **15** shows a thermal map **280** of results from thermal modeling performed on the door and frame segment assembly of FIG. **12**. Similar to the thermal map **260**, the steep temperature gradient on the interior side of the frame segment indicates that the backing member is preventing heat from the external environment from entering into the inside of the refrigerated enclosure. For illustrative purposes, the following are approximate temperatures at locations **282** through **299** shown in FIG. **15**: **282**: -9.7° F., **284**: 0.9° F., **285**: 1.5° F., **286**: 22.1° F., **288**: 43.2° F., **290**: 48.5° F., **292**: 53.8° F., **293**: 53° F., **294**: 59.1° F., **295**: 0.9° F., **296**: 11.5° F., **297**: 22.1° F., **298**: 32.7° F., **299**: 22.1° F.

In certain implementations, a frame includes an elongated edge on the front portion of the frame to increase heat absorption to keep temperature of the frame high enough to avoid condensation. In one implementation, the width of the forward flange of the main frame member is selected to increase heat absorption from the ambient warm air into the frame to inhibit condensation on the frame. An insulating strip may be included behind the forward flange (between the forward flange and the enclosure in which the frame is installed).

In some implementations, a temperature-controlled enclosure has a forward flange with physical characteristics that control thermal characteristics of a frame door interface. For example, an outer edge of the door can be kept above the dew point of ambient air.

Referring again to FIG. **7** main frame member **100** includes forward flange **108**. Forward flange **108** includes a front surface in contact with the ambient air outside of refrigerated enclosure **10**. Main frame member **100** can be made of a thermally conductive material, such as aluminum.

Forward flange **108** can absorb heat from the ambient air. A portion of the heat absorbed from the ambient air can be conducted through main frame member **100** (including middle wall **106** and base **102**) and transferred to contact plate **44**.

In some implementations, a length of a forward flange is selected to maintain temperatures in one or more locations of frame segment assembly **60**, door **36**, or both, in a target temperature range. As used herein, a “target” of a characteristic can be for a specific value or a range of values. A “target range” can have upper and lower bounds, or can be unlimited in one direction. For example, as one example, one target temperature range for a location on a contact plate can be 45 to 60 degrees F. Another target temperature range for a location on a contact plate can be 50 degrees F. and above.

In one implementation, a length *L* of forward flange **108** (measured from the top to bottom in FIG. **7**) is about 1.5 inches or more. In one implementation, the length of a forward flange is about 1.6 inches. In another implementation, the length of a forward flange is about 1.0 inches or more. In another implementation, the length of a forward flange is about 1.25 inches or more.

In some implementations, a surface area of a forward flange is selected to increase absorption of heat from ambient air. The front surface of forward flange can include, an arcuate shaped, ridges, grooves, bumps, or other members that increase a surface area in comparison to a flat surface. FIG. **16** illustrates a thermal frame having a forward ball flange. Frame segment member **300** includes ball flange **302**. Ball flange **302** includes front surface **304**. Front surface **304** of ball flange **302** includes rounded ridges **306** and corresponding grooves **308**. The undulations of front surface **304** result in a larger surface area of front surface **304**. In one implementation, the front surface area of a forward flange is about 1.5 inches or more per linear inch of frame segment. In another implementation, the front surface area of a forward flange is about 1.0 inches or more per linear inch of frame segment. In another implementation, the front surface area of a forward flange is about 1.25 inches or more per linear inch of frame segment.

FIG. **17** illustrates a thermal frame having a forward flange with a curved surface. Frame segment member **320** includes forward flange **322**. Forward flange **322** includes front surface **324**. Front surface **324** of forward flange **322** includes convex surface **326**. The convex surface **326** results in a larger surface area of front surface **324**.

FIG. **18** shows a thermal map **340** of results from thermal modeling performed on the door and frame segment assembly of FIG. **16**. As illustrated by the temperature regions extending along the outer member and to the contact plate, the thermally conductive forward flange of the frame assembly conducts heat from the external environment to the contact plate. Thus, the front portion of main frame member, the contact plate, and/or door can be maintained at or above a target temperatures in a manner such as described above. For illustrative purposes, the following are approximate temperatures at locations **341** through **351** shown in FIG. **14**: **341**: 1.1° F., **342**: 11.6° F., **343**: 27.5° F., **344**: 48.7° F., **345**: 44.7° F., **346**: -9.5° F., **347**: 1.1° F., **348**: 11.6° F., **349**: 27.5° F., **350**: 64.4° F., **351**: 27.5° F.

In various implementations described above, heat is absorbed from the ambient air in front of a refrigerated enclosure into a forward flange. In other implementations, thermally conductive segments or members of various shapes and forms can extend from, or be attached to, a thermal frame member to absorb heat from the ambient air

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in front of a refrigerated enclosure. For example, a U-channel or angle can be attached to forward flange 108. In some implementations, a front flange can include ribs, fins, corrugations or other features on the front of the flange.

In some implementations, condensation on a refrigerated enclosure is controlled by selecting one or more target thermal characteristics at one or more locations of a door/frame interface of a refrigerated enclosure. The method can include providing a thermal frame with a forward flange having one or more physical characteristics that maintain the at least one of the target characteristics. For example, the length of the forward flange can be selected to maintain a temperature at an outer edge of a door at or above a target temperature. In one implementation, the length of a forward flange is large enough to maintain an outer edge of a gasket for a door at or above a temperature at which condensation would occur.

In various implementations described above, a target thermal characteristic is a temperature or a temperature range. In some implementations, other thermal characteristics can be used. Examples of other thermal characteristics that can be used include a rate of temperature change, an amount of heat transfer, or a rate of heat transfer.

In certain implementations, a refrigerated enclosure includes a mullion having thickened sidewalk that reduce thermal transference from front to back of the mullion. Thermally insulating material (e.g., foam board) can be placed on the mullion sides. The mullion can include co-extruded portions, one of the co-extruded portions being of a lower density than the other co-extruded portion. The lower density material for the mullion may be, for example, a cellular material or ABS foam. The lower-density co-extruded portion is on the contact-plate side of the mullion. The lower-density co-extruded portion can receive a heater wire and zipper and serves as a thermal break.

In certain implementations, a refrigerated enclosure includes a mullion bracket that serves as a thermal barrier between the mullion and a frame segment to which the mullion is connected. The mullion includes a perimeter flange between the mullion and the frame. The bracket can restrict air from passing between the door frame and the mullion. A rectangular block of the mullion bracket can be inserted into a corresponding opening in the mullion. The block of the mullion bracket can be secured to the mullion by way of opposing fasteners in the lateral walls of the mullion.

In certain implementations, frame members, mullion members, or both, of a refrigerated enclosure have heater wire grooves that position a heater wire in direct contact with contact plate of the frame.

In various implementations described above, a bracket for a backing member includes a portion that is used to attach a main frame member to the body of an enclosure. In other implementations, a bracket for a backing member is separate from a bracket that is used to mount the main frame member to a body of an enclosure. In some implementations, a backing member may not include a bracket at all.

As used herein, “control” of a characteristic includes influencing or affecting a value of the characteristic. For example, an insulating member can be selected to control a temperature of a location on a frame member such that the temperature of a surface of a frame is maintained at a higher level.

As used herein, a “main” frame member includes any frame member to which other components of a frame

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assembly can be attached. For example, main frame member 100 provides a base to which contact plate 44 and backing member 102 can be coupled.

As used herein, a “member” can be a unitary structure or a combination of two or more members or components.

As used herein, “coupled” includes directly or indirectly connected. Two elements are coupled if they contact one another (e.g., where faces of a backing member and a contact plate are in contact with one another.)

As used herein, the terms “perpendicular,” “substantially perpendicular,” or “approximately perpendicular” refer to an orientation of two elements (e.g., lines, axes, planes, surfaces, walls, or components) with respect to one and other that forms a ninety degree (perpendicular) angle within acceptable engineering, machining, or measurement tolerances. For example, two surfaces can be considered orthogonal to each other if the angle between the surfaces is within an acceptable tolerance of ninety degrees (e.g., $\pm 1-5$ degrees).

It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

While a number of examples have been described for illustration purposes, the foregoing description is not intended to limit the scope of the invention, which is defined by the scope of the appended claims. There are and will be other examples and modifications within the scope of the following claims. For example, the construction and arrangement of the refrigerated case with thermal door frame as shown in the various exemplary embodiments is illustrative only. Although only a few embodiments of the present inventions have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the description and advantages of the subject matter disclosed herein. For example, elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present invention as defined in the appended claims. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present inventions.

What is claimed is:

1. A temperature-controlled enclosure for displaying cold items, comprising:

- a body comprising a front opening and defining an interior space of the temperature-controlled enclosure;
- a frame assembly coupled in the front opening of the body, wherein the frame assembly comprises a frame segment comprising, in cross-section:

- a thermally conductive main frame member comprising:

- a base comprising a rear wall and an interior wall;
 - and

- a middle wall coupled to the rear wall;

- a backing member coupled to the thermally conductive main frame member, the backing member comprising:

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a thermally insulative rear leg that couples on the rear wall of the base of the thermally conductive main frame member;

a thermally insulative interior leg that couples on the interior wall of the base of the thermally conductive main frame member; and

a bracket holding the thermally insulative rear leg and the thermally insulative interior leg to the thermally conductive main frame member, the bracket comprising a front surface; and

a contact plate coupled at a first end to the thermally conductive main frame member, and coupled at a second end to the front surface of the bracket; wherein the front surface of the bracket abuts the rear surface of the contact plate;

wherein the contact plate is in direct thermal contact with the thermally conductive main frame member at the first end and at a second location on the contact plate that is spaced apart from the first end; and

one or more doors coupled to the frame assembly, wherein at least one of the doors comprises:

one or more window panels; and

a gasket coupled on a rear surface of the door, wherein the gasket is configured to form a seal between the frame assembly and the door when the door is closed,

wherein a portion of the frame segment is configured to contact at least a portion of a rear surface of the gasket when the door is closed, and

wherein the middle wall extends forward at least to the one or more doors, the middle wall coupled to the contact plate.

2. The temperature-controlled enclosure of claim 1, wherein an inner surface of the backing member is interior to an inner surface of the gasket when the door is closed.

3. The temperature-controlled enclosure of claim 1, wherein an outer surface of the contact plate is substantially aligned with an inner surface of the gasket when the door is closed.

4. The temperature-controlled enclosure of claim 1, wherein a front surface of a clip attached to the backing member and a rear surface of the door define a channel interior to an inner surface of the gasket.

5. The temperature-controlled enclosure of claim 1, wherein the thermally conductive main frame member further comprises a forward flange coupled to the base by the middle wall, the forward flange configured to absorb heat from ambient air to inhibit condensation on the frame assembly.

6. The temperature-controlled enclosure of claim 5, further comprising an insulating member between the forward flange and a front surface of the body of the temperature-controlled enclosure.

7. The temperature-controlled enclosure of claim 5, wherein one end of the forward flange is coupled to the middle wall.

8. The temperature-controlled enclosure of claim 5, wherein the thermally conductive main frame member is coupled to the contact plate by the middle wall at a location between the forward flange and the rear wall.

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9. The temperature-controlled enclosure of claim 1, wherein the contact plate is flush in contact with the gasket.

10. The temperature-controlled enclosure of claim 1, wherein the gasket is positioned on the rear surface of the door to contact the contact plate when the door is closed.

11. The temperature-controlled enclosure of claim 1, wherein an outer face of the contact plate is flush with the gasket when the door is closed.

12. The temperature-controlled enclosure of claim 1, wherein an interior face of the bracket is offset from the gasket when the door is closed.

13. The temperature-controlled enclosure of claim 1, wherein the gasket, the door, and a clip attached to the bracket define a channel interior to the gasket when the door is closed.

14. The temperature-controlled enclosure of claim 1, wherein the thermally conductive main frame member holds one end of the contact plate to receive the gasket when the door is closed.

15. A frame for a refrigerated enclosure, comprising:

a frame segment comprising, in cross-section:

a thermally conductive main frame member comprising a base, the base comprising a rear wall and an interior wall;

a backing member coupled to the thermally conductive main frame member, the backing member comprising:

a rear leg that couples on the rear wall of the base of the thermally conductive main frame member; and

an interior leg that couples on the interior wall of the base of the thermally conductive main frame member, wherein the interior leg and the rear leg define a thermally insulating portion, wherein the thermally insulating portion forms an L-shape; and

a bracket holding the thermally insulating portion to the thermally conductive main frame member, the bracket comprising a front portion surface; and

a contact plate coupled at a first end to the thermally conductive main frame member, and coupled at a second end to the front surface of the bracket, wherein the contact plate extends inwardly over at least a portion of the front surface of the bracket, and wherein the contact plate is in direct thermal contact with the thermally conductive main frame member at the first end and at a second location on the contact plate that is spaced apart from the first end.

16. The frame of claim 15, wherein a middle wall of the thermally conductive main frame member comprises a slot configured to retain an edge of the contact plate.

17. The frame of claim 15, wherein the base comprises one or more channels configured to hold a heater wire, wherein at least one of the channels comprises an opening at least partially facing the contact plate.

18. The frame of claim 15, wherein the front surface is configured to extend forward to a rear plane of the contact plate.

19. The frame of claim 15, wherein the thermally insulating portion of the rear leg and the interior leg comprise a foam material.

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